

School of Engineering and Applied Science (SEAS), Ahmedabad University
CSE 400: Fundamentals of Probability in Computing
Lecture 6 Scribe: Discrete Random Variables & Expectation

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1 Random Variables – Motivation and Concept

A **Random Variable (RV)** is a function that assigns a real number to each outcome in the sample space.

Visualization of a Random Variable

The distribution of a random variable can be visualized as a bar diagram:

- The x-axis represents the values that the random variable can take.
- The height of the bar at value a_i is the probability $\Pr[X = a_i]$.
- Each probability $\Pr[X = a_i]$ is computed by evaluating the probability of the corresponding event in the sample space.

2 Discrete and Continuous Random Variables

2.1 Discrete Random Variable

A random variable is considered **discrete** if:

- It has countable support.
- It is described using a **Probability Mass Function (PMF)**.
- Probabilities are assigned to single values.
- Each possible value has strictly positive probability.

2.2 Continuous Random Variable

A random variable is considered **continuous** if:

- It has uncountable support.
- It is described using a **Probability Density Function (PDF)**.
- Probabilities are assigned to *intervals* of values.
- Each individual value has zero probability.

3 Example 1 – Tossing 3 Fair Coins

Problem Setup: The experiment consists of tossing 3 fair coins. Let Y denote the number of heads that appear. The possible values of Y are $Y \in \{0, 1, 2, 3\}$.

Computation of Probabilities

Each single outcome of 3 fair coin tosses (e.g., h, t, h) has a probability of $\frac{1}{8}$.

- $P(Y = 0)$: $\{t, t, t\}$

$$P(Y = 0) = \frac{1}{8}$$

- $P(Y = 1)$: $\{t, t, h\}, \{t, h, t\}, \{h, t, t\}$

$$P(Y = 1) = \frac{3}{8}$$

- $P(Y = 2)$: $\{t, h, h\}, \{h, t, h\}, \{h, h, t\}$

$$P(Y = 2) = \frac{3}{8}$$

- $P(Y = 3)$: $\{h, h, h\}$

$$P(Y = 3) = \frac{1}{8}$$

Verification of Total Probability

Since Y must take one of the values $\{0, 1, 2, 3\}$:

$$\sum_{i=0}^3 P(Y = i) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1$$

4 Probability Mass Function (PMF)

Definition: A random variable X is discrete if it takes on at most a countable number of possible values. Range: $R_X = \{x_1, x_2, x_3, \dots\}$ (finite or countably infinite).

The function $P_X(x_k) = P(X = x_k)$ for $k = 1, 2, 3, \dots$ is called the **Probability Mass Function (PMF)** of X .

Property:

$$\sum_{k=1}^{\infty} P_X(x_k) = 1$$

5 PMF – Worked Example

Problem: The PMF of a random variable X is given by:

$$p(i) = c \frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \dots \quad (\text{where } \lambda > 0)$$

Find $P(X = 0)$ and $P(X > 2)$.

Step 1: Determine the Constant c

Using the property that total probability sums to 1:

$$\sum_{i=0}^{\infty} p(i) = 1 \implies \sum_{i=0}^{\infty} c \frac{\lambda^i}{i!} = 1$$

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the Taylor series expansion $e^\lambda = \sum_{i=0}^{\infty} \frac{\lambda^i}{i!}$:

$$ce^\lambda = 1 \implies c = e^{-\lambda}$$

Step 2: Compute $P(X = 0)$

$$P(X = 0) = c \frac{\lambda^0}{0!} = e^{-\lambda} \cdot 1 = e^{-\lambda}$$

Step 3: Compute $P(X > 2)$

$$P(X > 2) = 1 - P(X \leq 2)$$

$$P(X \leq 2) = P(X = 0) + P(X = 1) + P(X = 2)$$

$$P(X \leq 2) = e^{-\lambda} + e^{-\lambda} \frac{\lambda^1}{1!} + e^{-\lambda} \frac{\lambda^2}{2!}$$

$$P(X > 2) = 1 - e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2} \right)$$

6 Bayes' Theorem – Recap

Using the definition of conditional probability $\Pr(A \cap B_i) = \Pr(B_i | A) \Pr(A)$, we obtain Bayes' Formula:

$$\Pr(B_i | A) = \frac{\Pr(A | B_i) \Pr(B_i)}{\sum_{j=1}^n \Pr(A | B_j) \Pr(B_j)}$$

- $\Pr(B_i)$: **A priori** probability
- $\Pr(B_i | A)$: **Posterior** probability

7 Bayes' Theorem – Example 1: Auditorium

Problem: An auditorium has 30 rows.

- Row k has $k + 10$ seats (Row 1 has 11, Row 30 has 40).
- A door prize is given by randomly selecting a row (equally likely, $1/30$), then a seat within that row.

Let S_{15} be the event that Seat 15 is chosen. Let R_k be the event Row k is chosen. Note: Seat 15 only exists in rows where total seats ≥ 15 . Since seats $= k + 10$, we need $k + 10 \geq 15 \implies k \geq 5$.

1. Computation of $P(S_{15})$

$$\begin{aligned} P(S_{15}) &= \sum_{k=5}^{30} P(S_{15} \mid R_k) P(R_k) \\ &= \sum_{k=5}^{30} \left(\frac{1}{k+10} \right) \cdot \frac{1}{30} \\ &= \frac{1}{30} \sum_{k=5}^{30} \frac{1}{k+10} \approx 0.0342 \end{aligned}$$

2. Conditional Probability $P(R_{20} \mid S_{15})$

Given that Seat 15 was picked, what is the probability it came from Row 20?

$$\begin{aligned} P(R_{20} \mid S_{15}) &= \frac{P(S_{15} \mid R_{20}) P(R_{20})}{P(S_{15})} \\ &= \frac{\left(\frac{1}{30}\right) \cdot \left(\frac{1}{30}\right)}{0.0342} \approx 0.0325 \end{aligned}$$

8 Bayes' Theorem – Example 2: Communication System

A communication system sends binary data (0 or 1). The receiver RX may make mistakes (0 detected as 1, or 1 detected as 0). The system is modeled using conditional probabilities of detection given the transmitted bit.

End of Lecture 6 Scribe